Pokhara University
Faculty of Science and Technology

Course Code.: COMP (3 Credits)	Full marks: 100
Course title: **IP Routing and Switching (3-1-2)**	Pass marks: 45
Nature of the course: Theory & Practice	Time per period: 1 hour
Year, Semester: VII	Total periods: 45
Level: Bachelor	Program: BE

1. **Course Description**
   Advanced Switching and Routing is a course that focuses on the design, configuration, and management of complex computer networks. The course covers advanced routing protocols, switching technologies, and network services used in enterprise and service-provider environments. Students learn how to build scalable, reliable, and secure networks by implementing dynamic routing, VLANs, inter-VLAN routing, redundancy, and Quality of Service (QoS). The course also emphasizes network optimization, troubleshooting, and best practices for high availability and performance in modern IP networks.
2. **General Objectives**
   Objectives of Advanced Switching and Routing:
   - Ensure **efficient and fast data packet forwarding**
   - Provide **scalability** for large and growing networks
   - Achieve **high availability and reliability**
   - Support **optimal path selection** using dynamic routing
   - Enable **load balancing** and congestion control
   - Improve **network security**
   - Support **Quality of Service (QoS)** for critical traffic
   - Optimize **network performance and resource utilization**
   - Simplify **network management and troubleshooting**
3. **Contents in Detail**

Specific Objectives	Contents
• Explain the role and importance of computer networks in modern society, business, and communication. • Describe how data is transmitted over a network, including basic concepts of data flow, transmission media, and communication models. • Identify and explain network protocol layering (OSI and TCP/IP models) and the function of each layer. • Understand network addressing concepts such as IP addressing, MAC addressing, and their roles in data delivery. • Analyze basic network requirements and design simple network topologies based on user needs. • Perform basic network configuration tasks and understand the purpose of common networking devices.	Unit I: Network Fundamentals 10 hrs 1.1 Living in a Network Centric World 1.2 Communication over the network, 1.3 Protocol Layers (OSI and TCP/IP models) and the function of each layer) 1.4 Addressing the Network such as IP addressing, MAC addressing 1.5 Planning and Configuring the Network.
• Explain the principles of routing, packet forwarding, and the role of routers in a network. • Configure and analyze static routes for simple and controlled network environments. • Explain the need for dynamic routing and compare different routing protocol types. • Describe the operation and characteristics of distance vector routing protocols. • Understand and implement RIP v1 routing in basic networks. • Design efficient IP addressing schemes using Variable Length Subnet Masking and Classless Inter-Domain Routing. • Explain improvements in RIP v2 and implement classless routing. • Describe EIGRP concepts, metrics, and advantages in enterprise networks. • Understand link-state routing principles and how routers maintain network topology. • Explain OSPF operation, areas, and basic configuration. • Describe BGP concepts and its role in inter-domain routing on the Internet. • Understand VPN concepts, types, and their role in secure communication. • Explain Layer 3 switching concepts and its role in high-performance routing.	Unit II: Routing and Packet Forwarding 15 hrs 2.1 Introduction to Routing and Packet Forwarding, 2.2 Static Routing, 2.3 Introduction to Dynamic Routing Protocols, 2.4 Distance Vector Routing Protocol, 2.5 RIP V1, VLSM and CIDR, 2.6 RIP V2, EIGRP, 2.7 Link State Routing, OSPF, BGP, Virtual Private Networks, Layer 3 Switching and Routing
• Describe switch operations, switching methods, MAC address tables, and basic switch functions. • Perform basic switch configuration and management tasks. • Explain the purpose of VLANs and configure VLANs to logically segment a network. • Describe VTP concepts and manage VLAN information across multiple switches. • Understand STP operation and its role in preventing switching loops in LANs. • Explain and configure inter-VLAN routing to enable communication between different VLANs. • Explain basic wireless standards, components, and operation.	Unit III: LAN Switching and Wireless 10 hrs 3.1 Introduction to LAN Design, 3.2 Basic Switch Concepts and Configurations, 3.3 Virtual LAN, VTP, 3.4 Spanning Tree Protocol, 3.5 Inter VLAN Routing, 3.6 Wireless Concepts and Configurations.

• Perform basic wireless configuration and understand wireless security fundamentals.	
• Explain the role and importance of computer networks in modern society, business, and communication. • Describe how data is transmitted over a network, including basic concepts of data flow, transmission media, and communication models. • Identify and explain network protocol layering (OSI and TCP/IP models) and the function of each layer. • Understand network addressing concepts such as IP addressing, MAC addressing, and their roles in data delivery. • Analyze basic network requirements and design simple network topologies based on user needs. • Perform basic network configuration tasks and understand the purpose of common networking devices.	Unit IV: Accessing the WAN 10 hrs 4.1 Introduction to WAN, 4.2 PPP, Frame Relay, 4.3 Network Security, Access Control Lists, 4.4 Virtual Private Networks, 4.5 IPV6 Addressing Services, 4.6 Network Troubleshooting, AAA

4. Methods of Instruction

Lecture, tutorials, lab works, projects.

5. List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course.

S.N.	Tutorials
1	Beginner-Level Projects: Design a basic enterprise network using **VLANs, inter-VLAN routing, and OSPF**
2	Intermediate-Level Projects: Build a **highly available campus network** with redundancy at both **Layer 2 and Layer 3**.
3	Advanced-Level Projects: Design a **large-scale enterprise WAN** using **BGP, OSPF, route redistribution, and security features**.

6. Practical Works

S.N.	Practical works
1	**Basic Network Configuration** Configure IP addressing, hostname, and basic device settings on routers and switches.
2	**Static Routing Configuration** Design a small network and configure static routes to enable end-to-end connectivity.
3	**RIP Version 2 Configuration** Configure and verify RIP v2 with classless routing.
4	**VLSM and CIDR Implementation** Design an IP addressing scheme using VLSM and configure routing accordingly.
5	**EIGRP Configuration** Configure EIGRP and verify routing tables, metrics, and convergence.

6	**OSPF Configuration** Configure single-area OSPF and verify neighbor relationships and routes.
7	**VLAN and Trunk Configuration** Create multiple VLANs and configure trunk links between switches.
8	**Inter-VLAN Routing** Configure inter-VLAN routing using a router-on-a-stick or Layer 3 switch.
9	**Access Control Lists (ACLs)** Configure standard and extended ACLs to control network traffic.
10	**Spanning Tree Protocol (STP)** Configure and analyze STP to prevent switching loops.

7. **Evaluation system and Students' Responsibilities**

Evaluation System

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class participation, quizzes, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

External Evaluation	Marks	Internal Evaluation	Marks
Semester-End Examination	50	Class attendance and participation	5
		Tutorials and projects works	5+5
		Quizzes/assignments and presentations	10
		Internal Term Exam	25
Total External	50	Total Internal	50
Full Marks 50+50 = 100			

Students' Responsibilities:

Each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

8. Prescribed Books and References

Text Book:

1. James F. Kurose, Keith W.Ross, "Computer Networking, A top down approach featuring the Internet "

Reference Books

2. A S Tanenbaum, "Computer Networks " 3. Behrouz A Forouzan, "Data Communications and Networking " 4. Todd Lammle, "Cisco Certified Network Associate Study Guide "